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1 Introduction

2 Background

2.1 Topology in geography

Topology is a mathematical discipline that “studies properties of spaces that are invariant to continuous deformation” (University of Waterloo (2024)). In geographic context, it means focusing strictly on the spatial relationships between features, such as connectivity, adjacency, and containment (citace), rather than their absolute geometric shape, distance, or magnitude. The discipline was formally established in the early 20th century, however its roots trace back to Leonhard Euler’s solution to the Seven Bridges of Königsberg problem (Euler (1741)). The objective of the problem was to cross every bridge in the city exactly once and return to the starting point. The physical distances and exact shapes of the landmasses were entirely irrelevant to the solution, the outcome depended solely on knowing which bridges connected which landmasses. By abstracting the city into a graph (see definition in chap. xx) while preserving the information about the connectivity of the landmasses, he was able to solve the problem.

-fig konigsberg bridges a graph

Probably one the most important connections between topology and geography is the Four Colour Theorem, proposed by Francis Guthrie 1852. It states that any planar map consisting of polygons can be coloured using only four colours without any pair of neighbouring polygons sharing the same colour. The theorem was proven more than a hundred years later by Appel and Haken (1977) and it relied entirely on topological principles, specifically, polygonal adjacency. Similarly to the Königsberg bridges, solving the problem required abstraction into a graph, a mathematical structure frequently used to describe spatial and topological relationships.

The formal study topological relationships is rooted in point-set topology, also known as general topology, as it is the foundation to many other related disciplines. It relies on the concepts set theory, introduced by can (1874), like set operations (e. g. intersection, union, complement), as well as the fundamental notions of subsets, open sets, and neighborhoods. Hausdorff (1914) formalized topological space in his book on set theory, introducing the

foundations of point-set topology.

A topological space is a pair (X, τ) , where X is a set of points and τ is a collection of subsets of X (the topology). τ must satisfy the following axioms:

1. Any union of elements of τ belongs to τ .
2. Any finite intersection of elements of τ belongs to τ .
3. X and $\emptyset$ belong to τ .

Based on the definition of topological space, Hausdorff (1914) formally defined the interior, boundary and exterior of a set. These three topological properties are the core of topological models used in spatial analysis. Given a set Y in topological space (X, τ) :

- interior (doplnim)
- boundary
- exterior

-Identifying these relationships is essential in numerous geographical disciplines, such as landscape ecology, hydrology, geography of transport, cartography and spatial statistics (Papadimitriou (2023)).

2.1.1 Topological models

During the 1980s, terms for spatial relations were mentioned in literature, however, they lacked formal definitions and were mostly treated as axiomatic (Egenhofer and Franzosa (1991)). The first rigorous framework for topological relationships was the *4-Intersection Model (4-IM)* developed by Egenhofer and Franzosa (1991). It applied interior and boundary definitions from point set topology to a geospatial context. The *4-IM* served as the predecessor to the *Dimensionally Extended 9-Intersection Model (DE9-IM)* (Clementini et al. (1993)), which is now the standard adopted by the *Open Geospatial Consortium (OGC)*. Expanding the *4-IM*, it is based on a 3×3 matrix of all possible intersections between interior, boundary and exterior of two geometries (point, line or polygon) and further describes the dimension of those intersections:

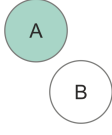
- False (-1) if there is no intersection,
- 0 if the intersection is a point,

- 1 if the intersection is a line,
- 2 if the intersection is a polygon.

-tab de9im matrix, nebo fig?

In most use cases, non-empty intersections (0, 1 or 2) can be universally treated as *True* when the result is invariant to the intersection dimension. However, for defining contiguities the dimension of the intersection is required (see chapt. xx). Specific value combinations in the matrix have standard names, which are referred to as spatial predicates. The *C++* library *GEOS* - a core dependency of many libraries (e. g. *Shapely*, *Geopandas*), databases (e. g. *PostGIS*), and applications (e. g. *QGIS*) - recognizes 9 spatial predicates (see tab ?) GEOS contributors (2025). For computational purposes, the matrix is often flattened into a 9-character DE-9IM string.

Table 1: Spatial predicates. - tohle taky dopnim

Spatial predicate	DE-9IM string	Description	Illustration?
Intersects	??		
Touches	FT***** F**T***** F***T****		
Disjoint	FF*FF*****		
Crosses	T*T***** 0*****		
Within	T*F**F***		
Contains	??		
Overlaps	T*T***T**		
Equals	TFFFTFFFT		
Covers	??		

Software libraries implement these predicates alongside additional ones as boolean functions, such as modified formulations (e. g. `contains_properly` in *Shapely* library)

or inverse relationships (e. g. `covered_by` in *Shapely*). Custom spatial queries can be formalized by defining the flattened DE9-IM string (e. g. `relate` in *Shapely*, `ST_relate` in *PostGIS*). Topological relationships derived from *DE-9IM* are the underlying framework for many spatial operations including overlay analysis, contiguity detection, and topology validation.

2.2 Topological validity and polygonal coverage

Spatial data often suffers from errors from a variety of sources. These inaccuracies are typically introduced during manual vectorization or data entry, data transfer, geometric editing operations (such as simplification or reprojection), or the integration of heterogeneous datasets of varying ages (Spatial Eye (2025), GIS Geography (2024), Servigne et al. (2000)).

Servigne et al. (2000) categorizes spatial consistency errors into three primary types:

- Structural errors: Arising when the chosen data structure cannot adequately accommodate the data model.
- Geometric errors: Conceptually invalid geometry (e. g. self-intersecting polygons).
- Topo-semantic errors: Occurring when the geometric representation violates real-world logical constraints (e.g., a single building footprint incorrectly intersecting two distinct parcels).

Geometric and topo-semantic errors are particularly problematic when detecting topological relationships, as they are not immediately noticable. The most commonly used data structures in modern GIS are “spaghetti” models (e.g., GeoJSON, Shapefile, Parquet, and GeoPackage). These models store each geometry as an isolated record with no information about their mutual relationships, therefore they need to be derived from the geometry (vertices). In presence of geometrical inaccuracies, accurate detection of topological relationships is impossible.

According to Ubeda and Egenhofer (1997), the topology error correction process consists of three steps: definition, checking, and correction. Definitions of the errors is essentially establishing the spatial constraints input data must satisfy. These user-defined constraints are derived from formal spatial predicates. Many of these rules are explicitly implemented

as ready-to-use topology checkers in desktop GIS software like *QGIS* and *ArcGIS Pro* (e. g. **must not have dangles** for line features, **must not overlap** for line or polygonal features).

Valid polygonal coverage requires specific set of topological constraints, which it must satisfy Jts (2022):

- polygon interiors must not overlap (no slivers),
- a shared boundary between polygons must have the same set of vertices (no gaps).

-geoplanar also mentions non-planar edges/touches.. planar enforcement

-If these constraints are violated, the logical contiguity between polygons is compromised. Enforcing valid polygonal coverage ensures accurate spatial analysis and improves cartographic visualization.

-fig topology errors slivers, gaps

Various software tools can evaluate the datasets for topology violations. *Shapely*, python wrapper for the *GEOS* library, implements functions for checking individual geometric validity rather than global topology. *JTS Topology Suite (JTS)*, an open source Java software library, can evaluate an entire polygonal coverage, returning error strings with the specific coordinates of violations. Desktop software like *QGIS* and *ArcGIS Pro* evaluate the selected topology rules (i. e. for polygonal coverage **must not overlap** and **must not have gaps**) and visually highlight every error on the map.

While *JTS*, *QGIS*, and *ArcGIS Pro* validate topology by checking individual geometries, Mapshaper temporarily converts the spaghetti model into an explicit topological data structure. Instead of storing independent geometries in rows, Mapshaper decomposes a polygonal dataset into an arc-node graph: creating edges (arcs) for boundaries and nodes where three or more polygons intersect. Polygons are then defined mathematically through sequences of these shared arcs, eliminating coordinate redundancy. In this explicit topological structure, the information detailing which polygons neighbor each other is inherently stored within the data model itself.

-Fixing errors can be either done (semi)automatically or by hand, involving a snapping threshold... a zapomněla jsem na geoplanar :(

-even though such tools exist, topological errors still occur, a presne proto pisu tuhle praci

2.3 Spatial weights and contiguity

Spatial weights, formalized by the seminal work by Cliff & Ord (1973), connect objects in geographic space using the spatial relationship between them. They serve for defining spatial neighbourhoods and geographically relevant sites. These relationships can be distance-based (k-nearest neighbours, distance bands), based on a membership in a geographic group (block weights) or adjacency-based (contiguity). Each pair of observations is assigned a weight, which can either be boolean (1 for relevance, 0 otherwise), or specific numerical value (e. g. inverse distance, hybrid weights). These spatial weights are usually stored in either spatial weights matrix or spatial weights graph, which will be explained in this chapter. Afterwards, we will further focus on contiguity-based spatial weights, as those can be directly impacted by violating polygonal coverage.

Spatial weight matrix, denoted as W , is one of structures used in the computation of spatial statistics. W is a $N \times N$ matrix, where N is the number observations. Each element w_{ij} represents weight between observations i and j , that is, how much observation j influences observation i . Because geographic datasets are rarely fully interconnected, most elements in W are 0. To ensure computational and memory efficiency, tools like libpysal (Rey et al. (2022)), geoda (Anselin et al. (2006)) or spdep (Bivand (2022)) also implement sparse matrix, which stores only non-zero elements.

Another structure used to store spatial weights is spatial weights graph. A weighted graph is defined as an ordered triple $G = (V, E, \omega)$, where V is a set of nodes, E is a set of edges representing connectivity between nodes and $\omega : E \rightarrow \mathbb{R}$ is a function that assigns real numbers (weights) to edges. Two nodes $i, j \in V$ are adjacent if there exists an edge $\{i, j\} \in E$ connecting them. Each node represents a polygon and a weighted edge denotes the relationship between observations.

-fig nejaky data, jejich spatial weight matrix, jejich graph

Fundamentally, spatial weights matrices and graphs store the exact same topological information; the matrix is simply the algebraic representation of the spatial graph.

Computational libraries accommodate both structures and allow for conversion between them (Rey et al. (2022), Bivand (2022)). While the matrix is the standard for calculating spatial autocorrelation formulas (Moran (1950), Geary (1954)), the graph provides more intuitive framework for clustering, where operations like minimum spanning trees and graph partitioning are necessary (Assunção et al. (2006), Duque et al. (2012)).

2.3.1 Contiguity

Spatial objects that share a common border are considered to be contiguous. However, “sharing a common border” isn’t a rigorous definition, leading to several formal interpretations of contiguity. The primary types of contiguity are named analogically to the movements of chess pieces: *Rook*, *Queen*, and *Bishop*.

For a pair of polygons to be considered *Rook* neighbours, the polygons must share an edge (a boundary of non-zero length). *Queen* contiguity is less strict - the polygons are considered neighbours if they share at least one point (vertex or a shared edge). *Queen* contiguity corresponds to the standard Touches spatial predicate. Bishop contiguity, rarely used on its own but occasionally a component of hybrid weights (Shortridge (2007)), is strictly defined as polygons sharing only a point, but no edges. Therefore, it represents the logical difference between *Queen* and *Rook* contiguity (*Queen* - *Rook*). All of these relationships can be formally expressed using the *DE-9IM* strings, as illustrated in Figure XX. Furthermore, these base definitions can be expanded into higher-order contiguity weights (neighbours of neighbours) to model relationships beyond immediate neighbours.

-fig 3 contiguity ilustrate s de9im stringy

-nonplanar enforced edges/touches geoplanar

-In geographical analysis, Rook or Queen contiguity is usually the standard. There is no consensus, which matrices use when, it is mostly experimental as it depends on the character of the data, but it impacts the outcome of the statistic.. nejaka hlubsi resers na specfikation of weight matrix?

-Fuzzy contiguity

-tady bude diagram pro premka

2.4 Spatial autocorrelation

2.4.1 Moran's I

2.4.2 Geary's C

2.5 Regionalization

2.5.1 Agglomerative clustering

2.5.2 SKATER

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